

# ASMIT DATTA

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## EDUCATION

### New York University

*M.S. in Computer Science*

New York, NY

Expected 2028

### Arizona State University

*B.S. in Computer Science, Minor in Business*

Tempe, AZ

May 2026

## EXPERIENCE

### Software Engineer Intern (Capstone)

*WatchDNA*

Aug 2025 – Apr 2026

Ontario, Canada (Remote)

- Built WatchDNA's store locator: an interactive Mapbox map with real-time filtering, an admin console for data operations, and an automated Python pipeline that discovers retailer endpoints without anyone pointing it at them
- Set up the backend from scratch on Node.js - the database schema, the REST endpoints, JWT authentication with role-based access, and the security middleware around it
- Shipped a companion app on the same infrastructure, reusing the auth, security, and data layers

### Software Engineer Intern

*MyYogaTeacher*

May 2025 – Jan 2026

San Diego, CA (Remote)

- Turned a high-level feature idea into diagrams, wireframes and user journeys for the development team, then built against them
- Shipped a voice-to-SQL pipeline on Whisper with streaming LLM responses that replaced the admin team's Ctrl+F-and-edit workflow
- Wrote the React Native client that streams biometric data in real time, with offline-first sync and time-series deduplication so nothing is lost when the network drops

### Technical Officer

*The AI Society at ASU*

Aug 2025 – Jan 2026

Tempe, AZ

- Mentored four freshmen through their first ML project, an emotion detection system reading facial expressions and audio cues
- Ran workshops and study sessions that made machine learning approachable to students with no technical background, and wrote the materials for them

### Research Study Assistant

*University of Delhi*

May 2024 – Aug 2024

New Delhi, India

- Worked under Prof. R. P. Singh on optimizing Gaussian Process classifiers for high-dimensional data

## PROJECTS

### Torii

[asmit.space/ky](https://asmit.space/ky)

2026

TypeScript, Bun, Hono, SQLite

- Put an MCP proxy between AI agents and the paid tools they call, enforcing identity, policy and budget on every request: JWT agent identity, a JSON policy engine, and a SQLite ledger recording every call and every reason one was blocked
- Handled the x402 payment flow end to end - intercept the HTTP 402, check the wallet, pay, retry - with a budget tracker that reserves funds before the upstream call so two agents cannot overspend at once; the whole thing ships on two production dependencies

### Lead Engine

[asmit.space/lead-engine](https://asmit.space/lead-engine)

2026

Python, FastAPI, React, SQLAlchemy

- Kept five scout agents running constantly across Zillow, Redfin, FSBO, expired listings and the Maricopa County Assessor, deduplicating by address and scoring every lead on motivation, ability and timeline with the reasoning written out - 2,129 leads sourced
- Built the dashboard over it as a live pipeline graph, so every scout, score and call is visible while it runs and a failed scout shows up the moment it fails

### Pash

[lnkd.in/p/dCwX6kYm](https://lnkd.in/p/dCwX6kYm)

2026

AWS S3/SQS, Demucs v4, CUDA

- Split songs into vocals, drums, bass and instrumentals with Demucs v4 - uploads land in S3, jobs queue through SQS, and separation runs on a remote CUDA worker, 5-10x faster than on CPU

### Portfolio

[asmit.space](https://asmit.space)

Three.js, Blender, React

- Modeled my own room in Blender and put it in the browser with Three.js - you walk up to the desk, click the machine, and every project opens as its own artifact instead of a list of links

## SKILLS

**Languages:** Python, Go, JavaScript, TypeScript, SQL, HTML/CSS, Java, Ruby

**AI & Agents:** Autonomous agents, Tool use, MCP, Prompt engineering, OpenAI API, Claude API, LLM streaming, x402, LangChain, Whisper

**Backend & APIs:** Node.js, REST, FastAPI, JWT, Express.js, Flask, Bun, Hono, GraphQL, WebSocket, Role-based access control, SQLAlchemy

**Data & Cloud:** AWS, PostgreSQL, SQLite, Docker, CI/CD, Automation, DevOps, Software testing, Git, Supabase, MongoDB, Firebase, CUDA

**Frontend & Mobile:** React, React Native, Expo, Tailwind, NativeWind, Material-UI, Three.js, Blender, Vite, Recharts, Electron, Figma, Mapbox